

Tyson Grover

E.I.T.

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- [Portfolio - TysonGrover.github.io](https://TysonGrover.github.io)

KEY SKILLS

- CAD: SolidWorks, OnShape, Fusion360
- FEA: SolidWorks, Ansys, Converge CFD
- Programming Languages: Python, MATLAB, C++, LabVIEW
- Manufacturing: CNC Mill, Manual Mill, Manual Lathe, 3D Printing
- Tools: Excel, Word, PowerPoint

EDUCATION

Bachelor's of Science, Mechanical Engineering

Aug '20 - May '26

Utah Tech University

Saint George, UT

- Notable classes: Autonomous Vehicles, Modern Controls, Thermal System Design, Computational Fluid Dynamics, Advanced Engineering Math
- GPA: 3.90

PROFESSIONAL EXPERIENCE

R&D Engineering Intern

Apr '22 - Present

RAM Aviation, Space, & Defense

Saint George, UT

- Developed precision control systems for hydraulic and pneumatic fluids at specified pressures and temperatures, successfully enhancing reliability and meeting all client requirements
- Enhanced temperature control accuracy and minimized solenoid valve errors by implementing a thermocouple-based PID controller with Arduino
- Improved solenoid valve cycling to sub-millisecond response using a single-board computer and LabVIEW programming, streamlining the testing process by 50%
- Engineered precision components with DFM/DFA principles and GD&T in SolidWorks, significantly increasing the efficiency of pressure and helium leakage testing tools
- Enhanced compliance and streamlined testing processes by meticulously analyzing ASTM, SAE, MIL, ASME, and ISO standards
- Assembled and tested 150+ valves sold to various aerospace companies

Manufacturing Teaching Assistant

Aug '21 - Apr '22

Utah Tech University, Department of Engineering

Saint George, UT

- Led & assisted in teaching a group of 20 students in basic manufacturing principles
- Provided training on manufacturing techniques such as milling, turning, and welding for introductory engineering projects, significantly enhancing student engagement through hands-on projects like a foldable knife

PROJECTS

Drop Shock Testing Tower

Aug '25 - May '26

- Creating pneumatic, electrical, braking and acceleration systems with Python and Raspberry Pi, contributing to a \$20K reduction in yearly expenses
- Used equations of motion & FEA to design tower to deliver up to 400 G's for 5 ms to a solenoid valve test unit
- Manufacturing to be done using waterjet, CNC mill, manual mill & lathe, laser welding

Boat Propulsion System

Jan '23 - May '23

- Constructed an innovative propulsion system for a paddleboat over a span of 4 months, significantly increasing public usability and enjoyment
- Engineered gear box and water-sealing enclosures with OnShape, enhancing design functionality through DFA/DFM principles

Volunteer

Missionary

Oct '23 - Jul '25

The Church of Jesus Christ of Latter-day Saints

Tempe, AZ

- Built deeper relationships and boosted individual confidence within faith-based groups through active listening, public speaking, and empathy
- Facilitated the successful execution of missionary initiatives using goal & plan setting, applying effective communication and leadership tactics to elevate performance metrics by 15%